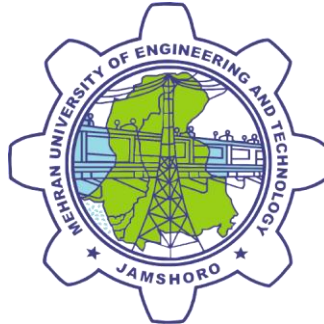


**MEHRAN UNIVERSITY OF ENGINEERING AND TECHNOLOGY,
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Department of Software Engineering

A SOFTWARE REQUIREMENT SPECIFICATION (SRS)
ON

Wind Turbine Performance Detection System

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Section 3

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SRS OF WIND TURBINE PERFORMANCE DETECTION SYSTEM

1. Introduction

This document serves as a comprehensive guide to the Software Requirement Specification (SRS) for the Wind turbine performance detection System. Its primary objective is to provide a thorough understanding of the project, its scope, objectives, and detailed requirements. By analyzing and documenting the essential features and functionalities of the project, this document aims to facilitate effective communication among stakeholders, guide the development team, and ensure the successful implementation of the system.

1.1 Purpose

- Provide graphical representations of wind turbine performance.
- Enable efficient error analysis and early warning detection from turbine sensors.
- Facilitate early detection of potential turbine failures through visual warnings.
- Maintain records of daily, weekly, and monthly electricity production for verification.
- Offer customizable graphical interfaces for comprehensive turbine monitoring.
- Ensure smooth data communication between sensors

1.2 Scope

The focused aim of the Wind turbine performance detection system is to detect any environmental, electrical and componential problems which can impact wind turbines physically including electrical sparks, and leakage of current, even though it destroys wind turbines. The system monitors these factors by effectively communicating with sensors installed on wind turbines that detect any outcomes and report them to software as a warning. Factors affecting Wind turbines can be, Earthquakes, High wind flow, and base foundation weaknesses of wind turbines. The system is designed to give real-time monitoring of Wind Turbines Electrical production, along with the day-to-day, month and weekly basis results in the form of Graphs and Reports as PDFs etc.

1.3 Overview

This document is a detailed overview of the Wind Turbine Performance Detection System Software. It explains the key components and functionalities of the software and is organized into several sections, each addressing specific aspects of the project.

The first section outlines the requirements of the Wind Turbine Performance Detection System, including hardware, users, and data and functionality needed to ensure its effective operation.

The second section provides a comprehensive description of the Wind Turbine Performance Detection System, highlighting its overall objectives, features, and scope. This section aims to give stakeholders a clear understanding of the software's purpose and capabilities.

Subsequent sections delve into the functional requirements, data requirements, design constraints, and underlying assumptions of the Wind Turbine Performance Detection System. These sections aim to define the system's functionalities, data processing needs, and

architectural considerations, laying the groundwork for its development and implementation.

1.4 Business Context

The Wind Turbine Performance Detection System Software has a profound impact not only on the operational efficiency, production output, and profitability of the company but also on its investors and collaborating third-party organizations. The effective performance of wind turbines directly influences the success of high-level projects, which have invested significant capital. The failure of even one turbine can disrupt production, leading to financial losses and the potential wastage or damage of essential turbine components. Therefore, ensuring the optimal performance of wind turbines is crucial for maximizing returns and maintaining positive partnerships within the industry."

2. System Description:

2.1 Product Function

The functionality of this system is that the different Controllers of the system can perform their respective roles efficiently and in an interactive way by integrating with system graphical representations.

Other Functionalities are already mentioned in the Functional Requirements section.

2.2 User characteristics

We have identified most of the users of this software in the Use case diagrams. The users that most frequently use the system are

Data Analyst: Analyze data, Access analyzed data and Generate reports.

Maintenance personnel: Access maintenance reports and receive maintenance alerts.

Other Users include a System Administrator, System Integrator and Wind turbine Operator.

2.3 Similar system information:

In exploring similar system info for the wind turbine performance detection system, we examine existing software solutions that offer comparable features and functionalities. By assessing similar systems, we gain valuable insights into industry best practices, emerging problems related to turbines and areas for innovation. This experience addresses us in designing a system that not only visualizes the graphical representation of the production and performance but also enables us to ensure that the communication interfaces plugged between different sources of information and alarming components work efficiently.

3. Functional Requirements

3.1 Functionality:

In this section, you will find the functional requirements for the Wind Turbine performance detection System(WTPDS).

3.2 Thumb Detection for Login Purposes:

This factor adds layers of additional security and ease to the system. The Users (Maintenance personnel, System Administrator, Data Analyst, System Integrator and Wind Turbine Operator) can now log in with advanced biometric authentication by thumb.

3.3 Graphical Performance Representation:

The monitored information for Wind Turbines is represented graphically by introducing graphs such as line graphs, pie charts, pictograph etc. The System Controllers can view performance trends and anomalies and summarize them with interactive charts, graphs and visualizations through different and multiple representation interfaces. The production performance can be if needed day to day, monthly and also for a week. By introducing these filters one can get the idea about the best performance of the wind turbines.

3.4 Granular Turbine Selection:

The System should allow the Controller to select the geographic areas or predefined zones such as northeast, southwest, etc. For the monitoring purposes for a specified area. Such as if wind turbine area(#b7-north-west) has encountered a warning or issue among one turbine then, the controllers can efficiently navigate to the area and then the specified turbine of that pointing area. This enables geographically targeted analysis and monitoring. The navigation functionality introduces a friendly interface as a map for navigation purposes.

3.5 Data Reporting and Analysis:

The system is designed to produce comprehensive reports that summarize the performance of wind turbines in terms of energy generation, efficiency, downtime, and maintenance alerts. Users will have access to customizable report templates and scheduling options to generate periodic reports. Additionally, the system is equipped with export capabilities, enabling users to save reports in various formats such as PDF and CSV for offline analysis or sharing.

3.6 System Administration:

To ensure the efficient management of user accounts and system settings, the system must provide robust administrative features. These features must enable administrators to easily and precisely configure authentication methods, access controls, and system preferences. Additionally, audit logs and activity tracking functionalities must be in place to enable real-time monitoring of user interactions and system events. Failure to provide such features could lead to security breaches, data loss, and other serious issues that can compromise the integrity of the system. Therefore, the system you use must have these features to prevent any mishaps.

3.7 Effective Communication with Sensors:

The software shall establish reliable communication channels with various sensors installed on wind turbines. It shall support protocols for data transmission and reception from sensors, ensuring seamless integration and data exchange.

3.8 Integration with Data Collection Systems:

The system will integrate with various data collection systems to gather and aggregate information. It will facilitate smooth synchronization and aggregation of data from different systems, ensuring comprehensive and accurate analysis. The software will provide user-friendly

functionalities for monitoring and analyzing data, enabling users to make informed decisions based on the insights gained.

3.9 Defining Protocols and Standards:

The software shall establish communication protocols, formats, and standards to ensure seamless data exchange. It shall select compatible protocols (e.g., MQTT, RESTful APIs) and data formats (e.g., JSON, XML) for consistency.

3.10 Compatibility with Existing Software Platforms:

Your software will agree with existing software platforms used by the wind turbine company. This ensures interoperability and slick integration with the company's existing infrastructure, minimizing disturbances and maximizing efficiency.

4. Non-Functional Requirements

4.1 Performance Requirements:

The system needs to be capable of processing data from wind turbines in real-time to provide timely insights into their performance. The graphical representation and data reporting features must have minimal response times to ensure that users can access information promptly. Additionally, the system should be scalable to accommodate increasing data volumes and user requests as the number of monitored turbines grows.

4.2 Reliability:

The system needs to guarantee that the performance data shown and reported to users is accurate and reliable. It should also be able to handle errors or faults gracefully without affecting the overall functionality of the system. To ensure continuous operation and minimize the risk of system downtime, redundant components should be implemented.

4.3 Availability:

The system must prioritize high availability to ensure that users can access it whenever necessary. It is essential to implement data backup and recovery mechanisms to prevent data loss in case of system failures or natural disasters. Distributing the workload evenly across system components is crucial to prevent overloading and ensure consistent performance.

4.4 Usability:

Usability is all about how user-friendly a system is and how easily users can interact with it. The system's user interfaces should be designed in an intuitive way, making it easy for users to navigate through turbine selection, view performance graphs, and generate reports. The login process, especially the thumb detection login and other authentication procedures, should be user-friendly and efficient.

4.5 Control or Security Requirements:

The security layer for login purposes should be implemented for the authorization mechanism to make sure that the Controllers only can access sensitive information and perform actions. Sensitive data stored in the system should be encrypted to prevent unauthorized access or interception.

4.6 Error Handling:

The WTPDS system should be equipped with effective error handling mechanisms to detect any issues related to data transmission, turbine selection, or any other operations. In case of errors, users should receive clear and informative error messages to guide them on how to resolve the issue or recover from it. To minimize any disruptions, the system should gracefully recover from errors. It is advisable to implement logging and monitoring functionalities to track system errors and performance issues for troubleshooting and analysis purposes.

4.7 Efficiency:

The system must have the capability to communicate efficiently with sensors, gather and process data, and create reports. It is imperative to optimize resource utilization, including CPU and memory, to ensure that the system functions seamlessly, even during times of high demand. The system resources must be optimized to reduce resource consumption and enhance efficiency, guaranteeing optimal performance even under heavy workloads. To process vast datasets and complicated calculations effectively, develop algorithms and data processing methods that are scalable and efficient.

5. Design Constraints:

5.1 Algorithmics Constraints:

Dictionary learning is a type of unsupervised machine learning used in signal processing. It allows for the creation of a set of features that are specific to a particular signal. These features can be used to monitor the condition of rotating machines, such as wind turbines. One of these features is called dictionary distance, and its effectiveness is dependent on selecting the appropriate hyperparameters for dictionary learning and the availability of data. However, data availability can be limited in condition monitoring systems for wind farms that are located remotely.

ANN: Detection and isolation technique for sensor calibration drifts on the purpose of measurement validation was developed. Temperature sensor measurements from the Supervisory Control and Data Acquisition system of a wind turbine were used for this aim. Low output rate of the measurements and nonlinear characteristics of the system drive the necessity to design an advanced fault detection algorithm. Artificial neural networks were chosen for this purpose considering their high performance in nonlinear environments. The results demonstrate that the proposed method can effectively detect the existence of calibration drift and isolate the exact sensor with faulty behavior.

5.2 Hardware Constraints:

➤ Graphics Processing Unit:

To boost up the neural network computations and measurement, a dedicated GPU is recommended. GPUs with CUDA support properly speed up training and inference tasks, so improving the overall efficiency of the system.

➤ Storage:

Storage capacity is necessary for storing overall data, system logs and files. SSD (Solid-state drive) can read/write fast is preferable to ensure quick access to data and improve overall system responsiveness.

➤ Memory:

A minimum of 8GB Ram to support smooth operation and prevent performance bottlenecks is recommended.

➤ Processing Power:

Due to too much complex controlling and dealing of software functionalities a robust processing unit capable of handling a high level computations involved in neural network training and real time data processing. Multicore processors with significantly high clock speed are recommended to ensure optimal performance.

➤ Environmental Considerations:

Hardware parts that should be capable of operating reliably in the environmental conditions typical of wind turbine installations. This includes

- Platinum resistance temperature sensors.
- Accelerometers.
- Ultraprecise temperature sensors.
- Speed sensors.

6. Project Requirements:

6.1 Specific Requirements:

→ Functional Requirements

We have already covered it in the section 3.1 to 3.11 enlisting functionalities like Graphical Representation, data analysis, sensor communication and more.

→ Non Functional Requirements

We have already covered it in the section 4.1 to 4.7 which address aspects like security, usability, performance.

6.2 Overall Project Goals:

The focused aim of the Wind turbine performance detection system is to detect any

environmental, electrical and componential problems which can impact wind turbines physically including electrical sparks, and leakage of current, even though it destroys wind turbines. The system monitors these factors by effectively communicating with sensors installed on wind turbines that detect any outcomes and report them to software as a warning. Factors affecting Wind turbines can be, Earthquakes, High wind flow, and base foundation weaknesses of wind turbines. The system is designed to give real-time monitoring of Wind Turbines Electrical production, along with the day-to-day, month and weekly basis results in the form of Graphs and Reports as PDFs etc.

6.3 Success Criteria:

- Reduced maintenance cost due to early detection of faults.
- Early detection of Potential failure.
- Early detection is key to avoiding catastrophic system failure.
- Efficient navigation and understanding of production information, wind turbine performance.

7. Other Requirements

7.1 Budget:

S.no	Task	Estimated cost	Description	Timeline (weeks)	Resources Required
1	Requirements Gathering				
	Stakeholder Interviews	\$2,000	Conducting interviews with stakeholders to gather specific requirements.	2	Business Analyst
	User Research	\$1,500	User surveys and interviews to understand user needs.	3	UX Researcher
2	UI/UX Design				
	Wireframing	\$1,200	Create wireframes to visualize the layout and structure.	2	UI/UX Designer

	Mockups	\$1,500	Develop mockups to refine the visual design.	2	UI/UX Designer
3	Software Development				
	Frontend Development	\$5,000	Develop frontend components and user interface	6	Frontend Developer
	Backend Development	\$7,000	Build backend systems and database architecture	8	Backend Developer
4	Testing and QA				
	Test Planning	\$1,500	Develop test plans to ensure comprehensive test coverage.	2	QA Engineer
	Test Execution	\$2,500	Execute tests and validate system functionality	4	QA Engineer
	Bug Fixing	\$1,000	Addredd and resolve identified bug and issues.	2	Development Team
5	Development and Launch	\$10,000	Deploy the system to production environment.	4	DevOps Engineer
6	Maintenance and Support				
	Bug Fixing and Updates	\$3,000	Provide ongoing maintenance and updates to the system.	Ongoing	Development Team
	Technical Support	\$4,000	Offer technical assistance and troubleshooting to users.	Ongoing	Support Team
7	Contingency	\$5,000	Allocate reserve funds for unforeseen expenses or changes	-	-
	Total	\$45,200			

8. Acronyms:

1. WTPDS: Wind Turbine Performance Detection System.
2. PDF: Portable Document Format.
3. DFD: Data Flow Diagram.
4. SSD: Solid State Drive.
5. XML: Extensible Markup Language.
6. API: Application Programming Interface.
7. CUDA: Compute Unified Device Architecture.
8. RAM: Random Access Memory.
9. MQTT: Message Queuing Telemetry Transport.
10. JSON: JavaScript Object Notation.
11. ANN: Artificial Neural Network.

9. References:

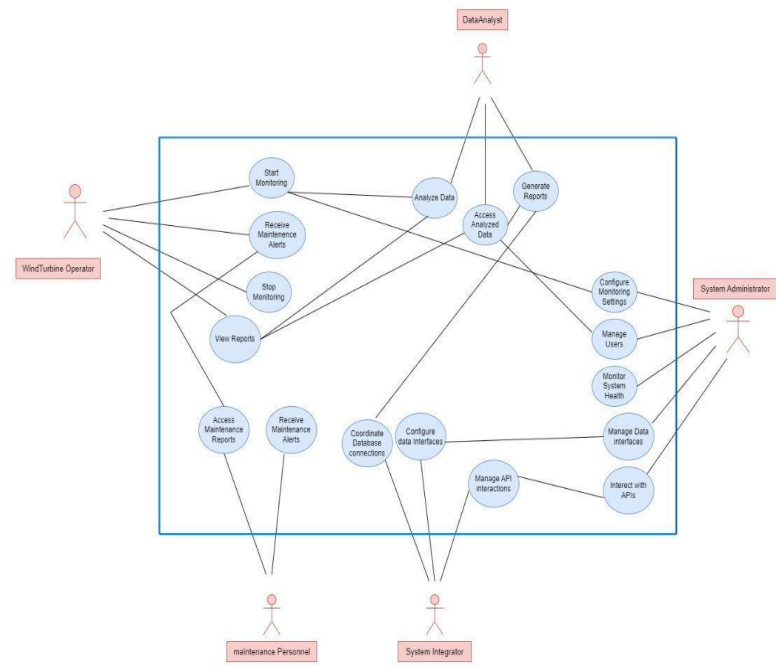
Enhancing Wind Turbine Performance: Statistical Detection of Sensor Faults Based on Improved Dynamic Independent Component Analysis.

→ <https://www.mdpi.com/1996-1073/16/15/5793>

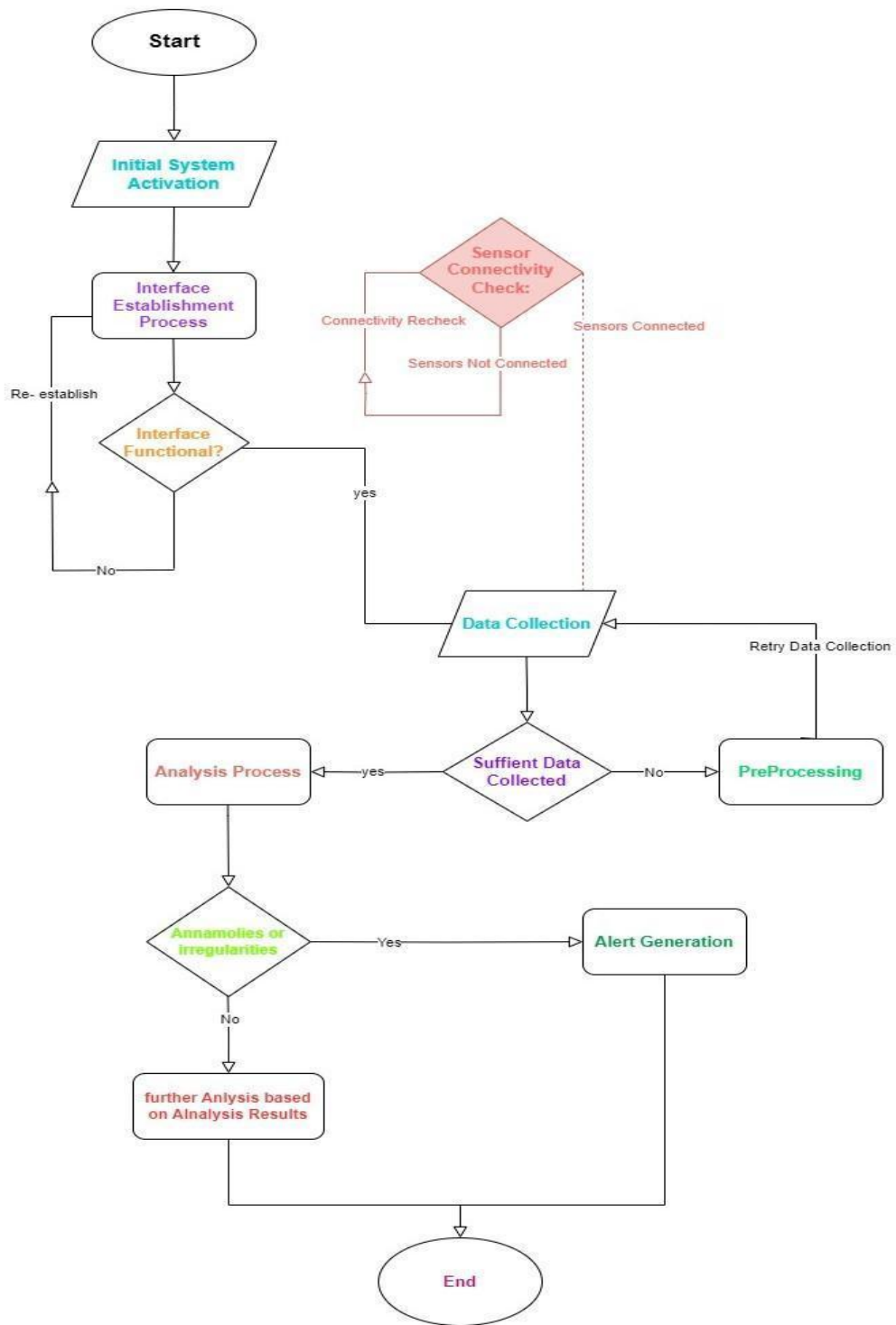
Home Page:

<https://taiyababhathi.github.io/Wind-Turbine-Project/>

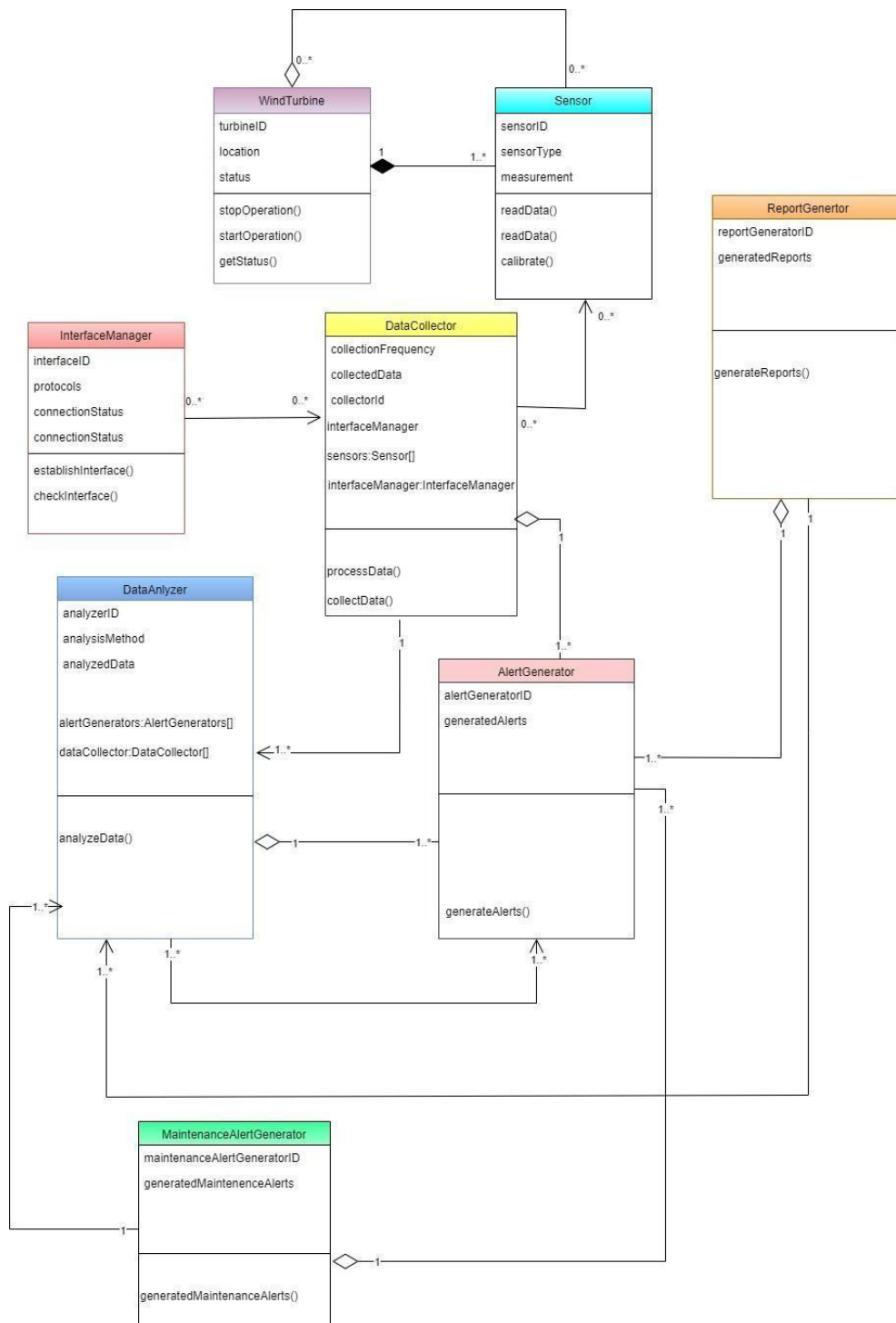
10. Diagrams:



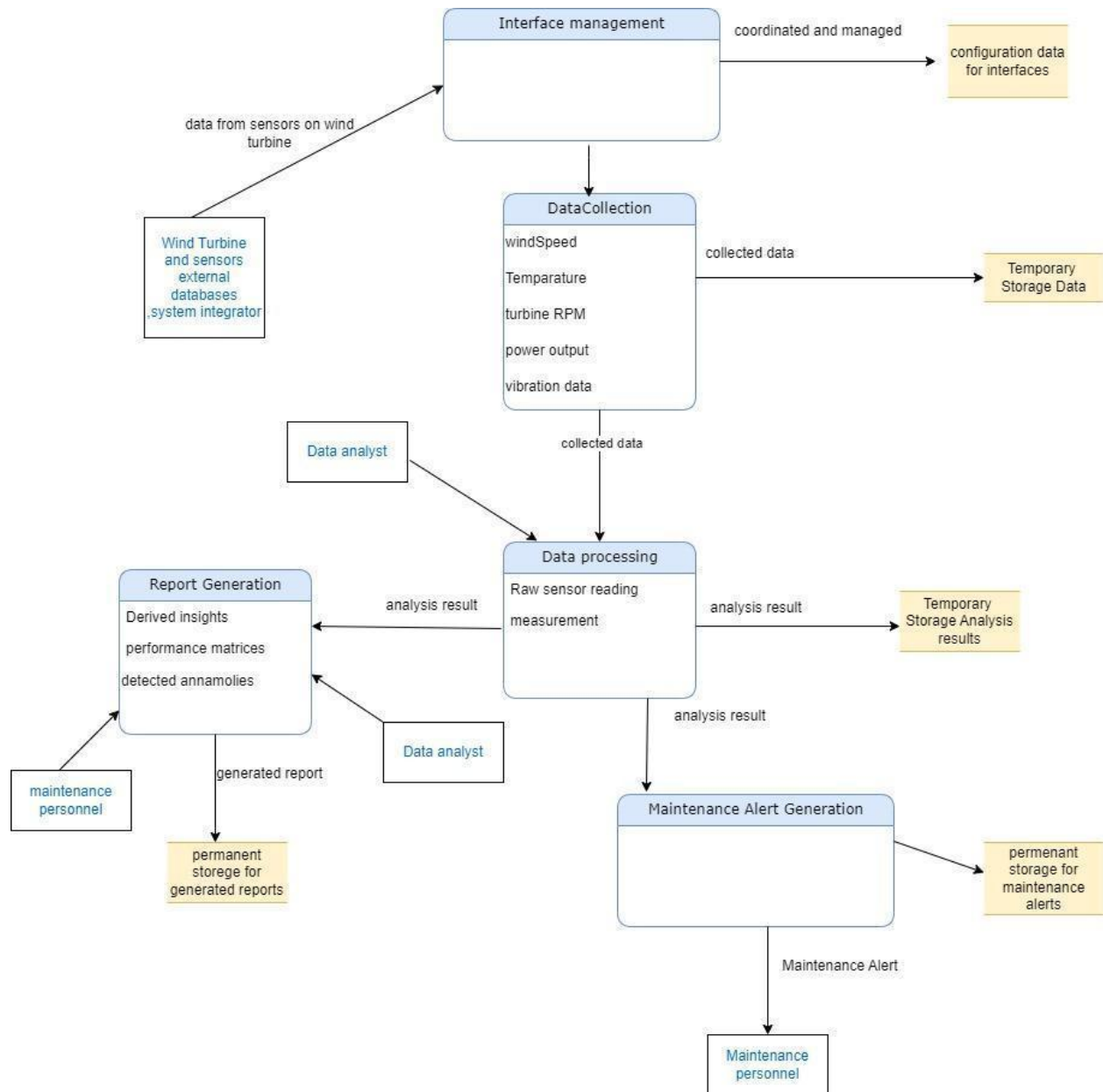
UseCase Diagram Of WTPDS



FlowChart Diagram Of WTPDS



Class Diagram Of WTPDS



DataFlow Diagram Of WTPDS